Pratyaksha Gupta

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Education

Ajay Kumar Garg Engineering College (AKGEC) Ghaziabad, UP

B.Tech in Information Technology *2023 – 2027*

- CGPA: 8.2 / 10
- Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Python Programming, Statistical Modeling, Computer Networking, Cloud Computing

Experience

Robotics Club, AKGEC Ghaziabad, UP

AI & Automation Programmer *2023 – Present*

- Developed and deployed AI/ML pipelines for real-time perception, decision-making, and autonomous control in competitive robotic systems
- Led technical implementation of sensor fusion, computer vision, and intelligent path-planning modules for national-level competition projects
- Built automation logic for task coordination and cross-functional system integration across hardware and software teams

Projects

VIGIL – AI-Powered Industrial Safety Intelligence System | *ET AI Hackathon 2.0 Semifinalist* 2026

- Architected a 9-module multi-agent safety monitoring system using Python threading and synchronous callbacks to detect fatal multi-factor industrial overlaps (gas leaks + active work permits + shift changes)
- Engineered an 11-rule heuristic Compound Risk Engine (M3) with an automated 0–100 risk-scoring system and a RAG memory pipeline (FAISS + Claude API) to retrieve past incidents and cite regulatory standards (OISD, Factory Act)
- Implemented real-time computer vision (M9) with OpenCV and optional YOLOv8 for PPE compliance and worker-presence tracking on edge infrastructure
- *Tech Stack:* Python, Streamlit, FAISS, OpenCV, YOLOv8, Pandas, Plotly, Claude API

GUJCOST Robofest 5.0 – Autonomous Drone Swarm | *1st Place, 10L Prize* Aug 2025 – Mar 2026

- Architected an AI-powered multi-drone system integrating computer vision, real-time hazard detection, and autonomous path planning
- Designed and trained ML models for object classification and obstacle avoidance in dynamic, unstructured environments
- Built a real-time multi-agent coordination framework enabling synchronized task execution across the drone swarm; filed patent (application in process) for the underlying mine-detection navigation system
- *Tech Stack:* Python, OpenCV, Scikit-learn, Raspberry Pi

ClauseLogic – GenAI Document Analysis System 2025

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline using LangChain, a FAISS vector store, and LLM APIs for intelligent document Q&A
- Implemented semantic chunking, embedding generation, and vector similarity search over unstructured legal/business documents
- Automated structured insight extraction via prompt engineering and deployed a full-stack app with a FastAPI backend and Next.js frontend
- *Tech Stack:* Python, LangChain, FAISS, OpenAI API, FastAPI, Next.js

ROBOCON 2025 – Autonomous Robot Control System | *All India Rank 13* Oct 2024 – Jul 2025

- Developed autonomous decision-making and motion-planning software integrating computer vision with real-time sensor data for context-aware task execution
- Designed real-time sensor fusion algorithms and programmed embedded feedback loops across actuation and perception layers; wrote and optimized 1000+ lines of embedded C++
- *Tech Stack:* C++, Python, Arduino, ESP32, OpenCV

Achievements

ET AI Hackathon 2.0 Semifinalist (2026): VIGIL – AI-powered industrial safety intelligence system
1st Place – GUJCOST Robofest 5.0 (2026): Autonomous drone swarm system, 10L prize
All India Rank 13 – ABU Robocon 2025: National-level autonomous robotics competition
Patent Filed (In Process): Autonomous drone swarm for mine detection and safe navigation
3rd Place – IIT Roorkee: National-level on-the-spot storytelling competition (10,000 prize)
Certification: Google Data Analytics Foundations – Coursera

Technical Skills

Languages: Python, C++, SQL, JavaScript (basic)
AI / ML: Scikit-learn, OpenCV, YOLOv8, NumPy, Pandas, Matplotlib, Seaborn, Plotly, TensorFlow, Prompt Engineering
GenAI & Agentic AI: LangChain, FAISS, Retrieval-Augmented Generation (RAG), OpenAI API, Claude API, Streamlit
Automation & Backend: FastAPI, REST APIs, Workflow Automation, Process Optimization
Databases: MySQL, FAISS (vector search)
Embedded & Robotics: Raspberry Pi, Arduino, ESP32, Sensor Integration, Real-Time Systems
Concepts: Machine Learning, Computer Vision, NLP, Autonomous Systems, Vector Search, Data Pipelines, Statistical Modeling, Intelligent Automation
Tools: Git, GitHub, VS Code, Arduino IDE